

Computation on Evolving Substrates

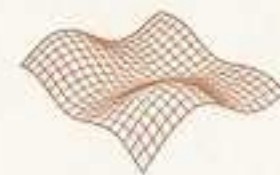
Dynamic Admissibility
and the Path-Dependence
of Constraint Geometry

> Why the 'stage' of computation
is actually an actor.

> CONSTRAINED SPACE

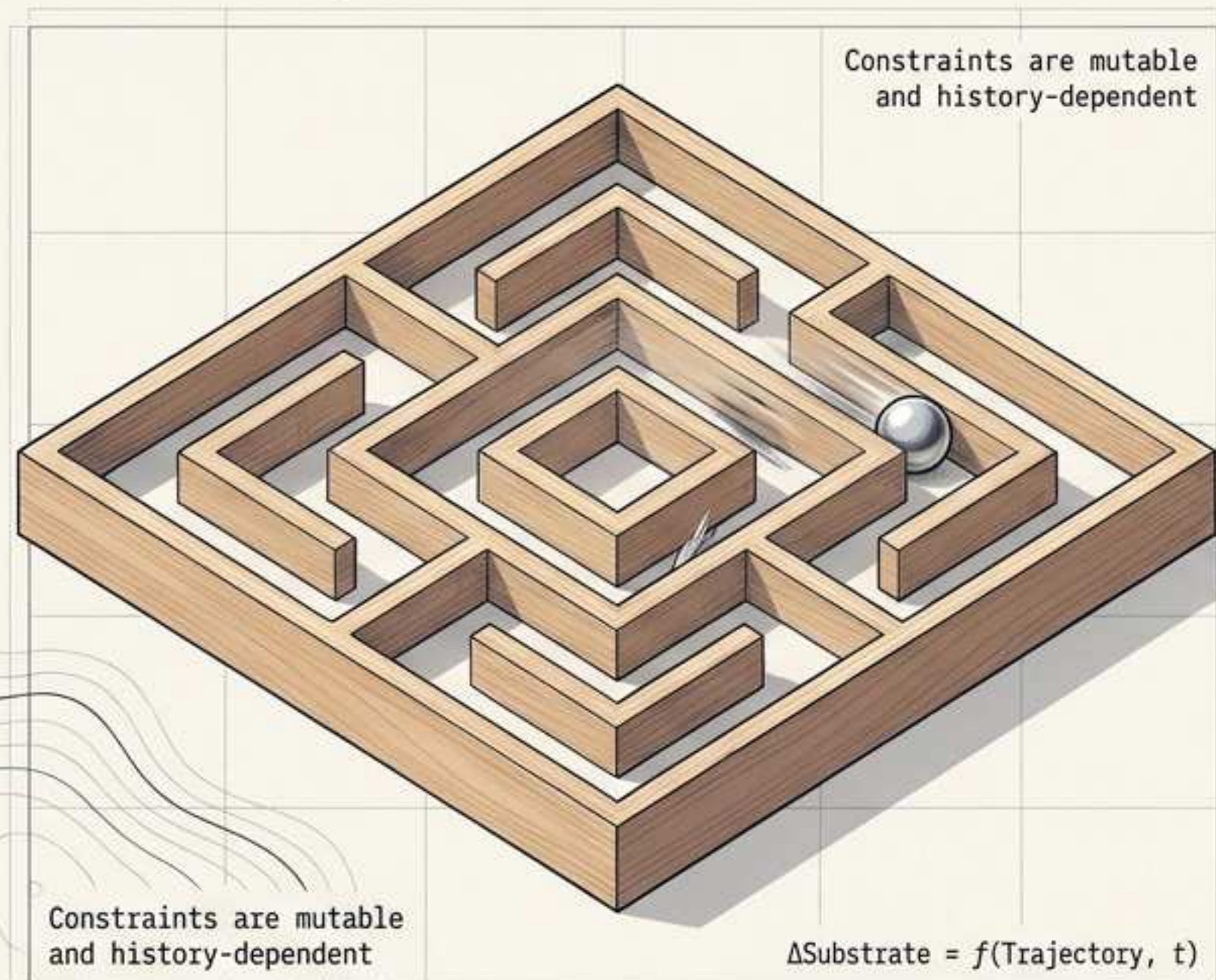


> DYNAMIC MANIFOLD



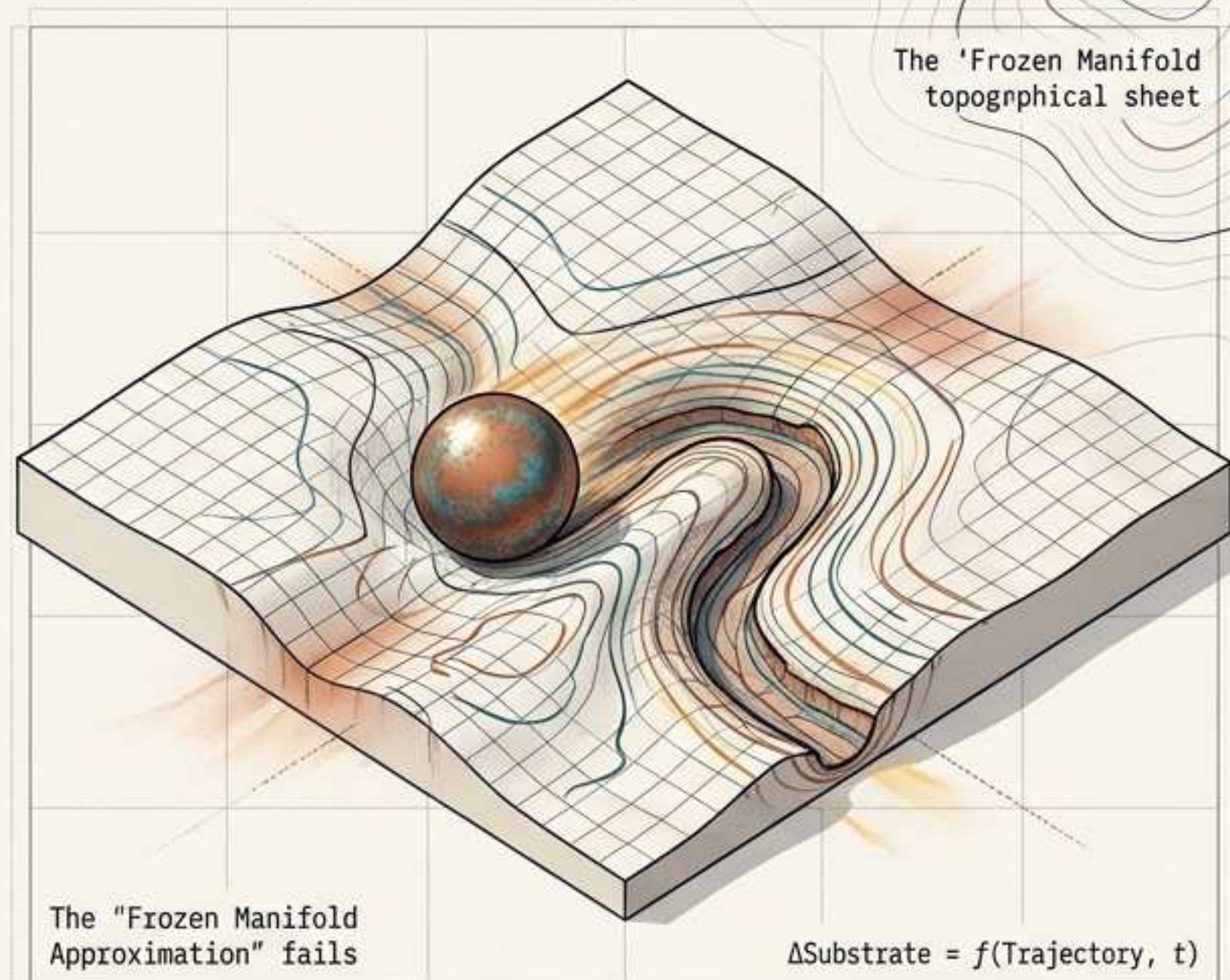
The Core Illusion: The Frozen Manifold

The Fixed Stage



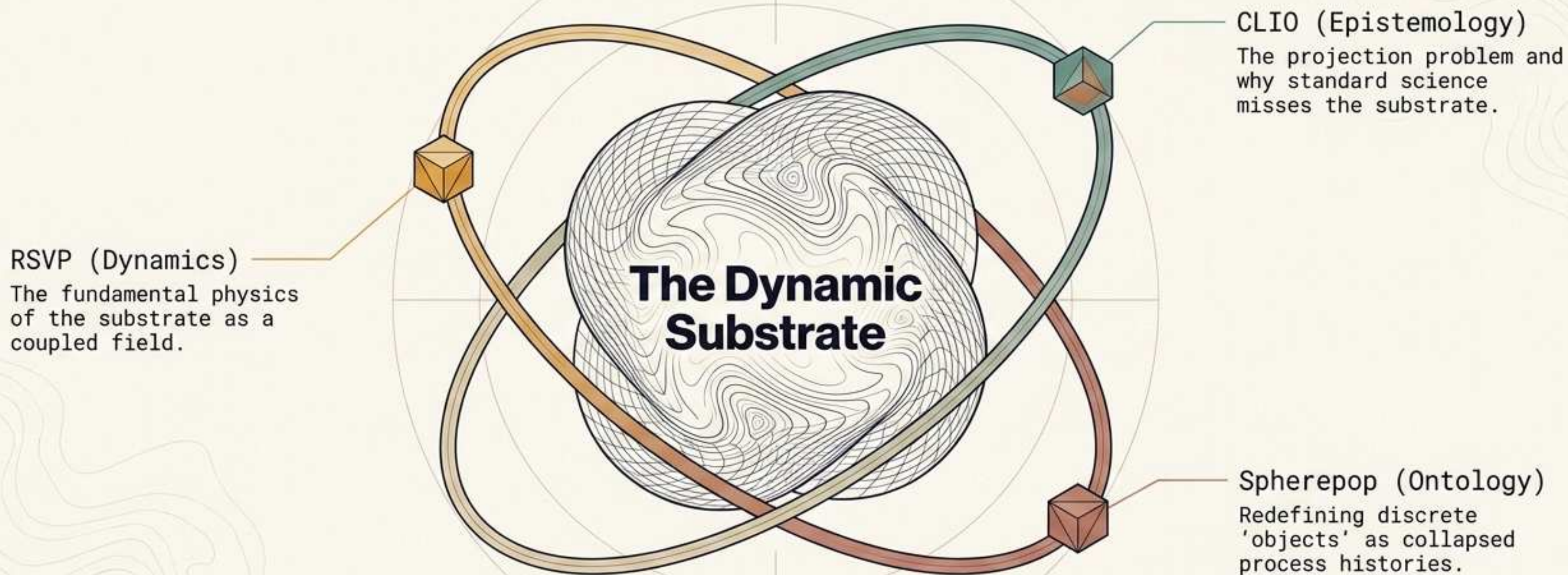
Standard models in science, economics, and AI assume a fixed constraint geometry. They track the trajectories of actors but ignore the substrate they move upon.

The Path-Dependent Reality



The "Frozen Manifold Approximation" fails whenever the medium is modified by the trajectories passing through it. Admissibility is not given; it is path-dependent.

The Paradigm Shift: Dynamic-Substrate Computation



Computation is determined not merely by trajectories, but by a substrate continuously reshaped by those trajectories.

Pillar 1: RSVP (The Relativistic Scalar-Vector Plenum)

The substrate is not empty space; it is a coupled field.

S: Dissipation

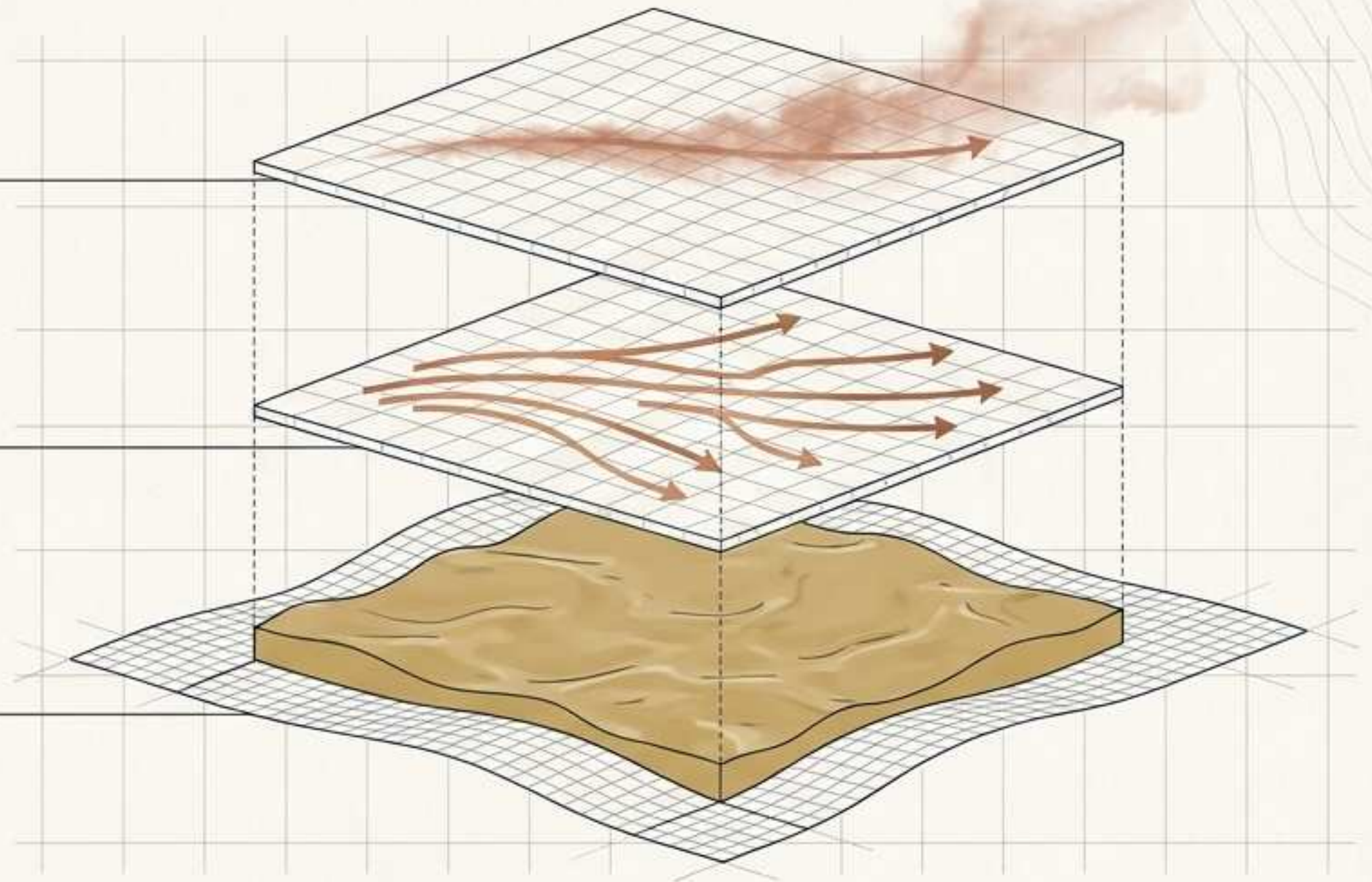
Irreversible burden, constraint accumulation, and metabolic cost.

v: Transport

Directed flow of influence and signal propagation velocity.

Φ : Capacity

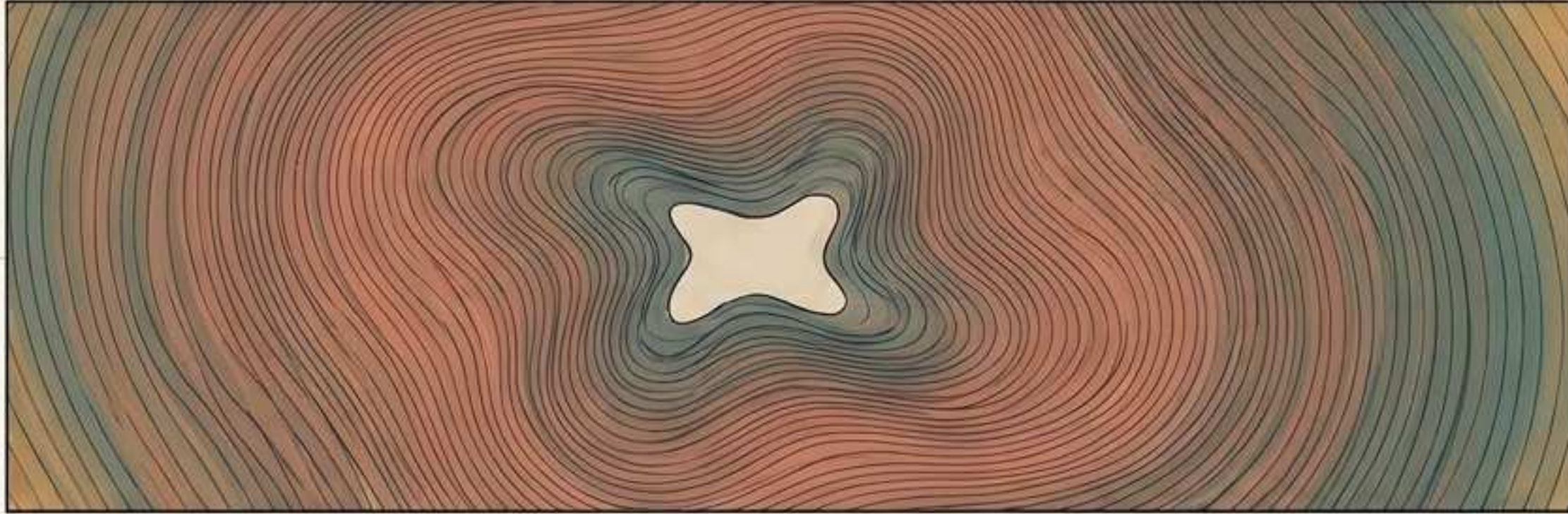
Local availability (e.g., ionic potential, economic resources).



$$\partial_t \Phi + \nabla \cdot (\Phi \mathbf{v}) = -\alpha S + R\Phi \quad (\text{Capacity is consumed by dissipation and restored by regeneration})$$

The Mechanics of Admissibility

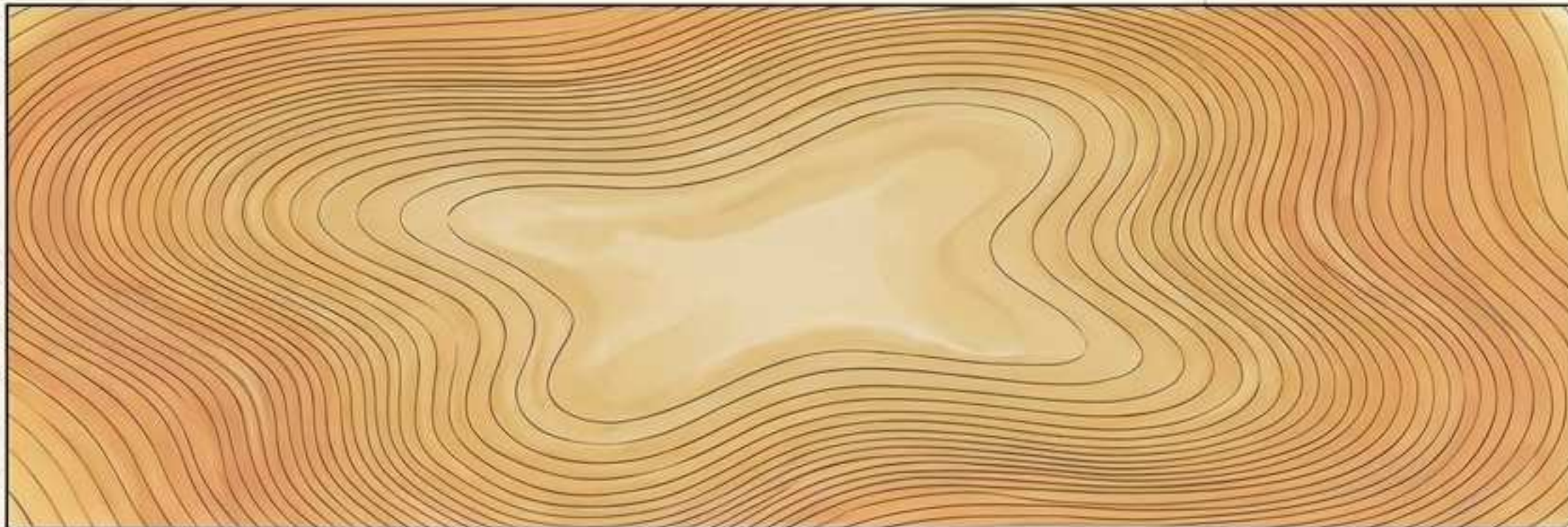
Admissibility is path-dependent. Without regeneration, systems exhaust their behavioral repertoire.



Wakefulness (Contracting)

Fast neural activity generates net dissipation (S).

The **Admissible Region**—the space of possible actions—shrinks.



Sleep (Expanding)

Substrate maintenance ($R\Phi$) clears the accumulated dissipation.

The **Admissible Region** expands, restoring reachable future states.

Pillar 2: CLIO (The Projection Problem)

The substrate variables are rendered invisible by standard projections.

The Error

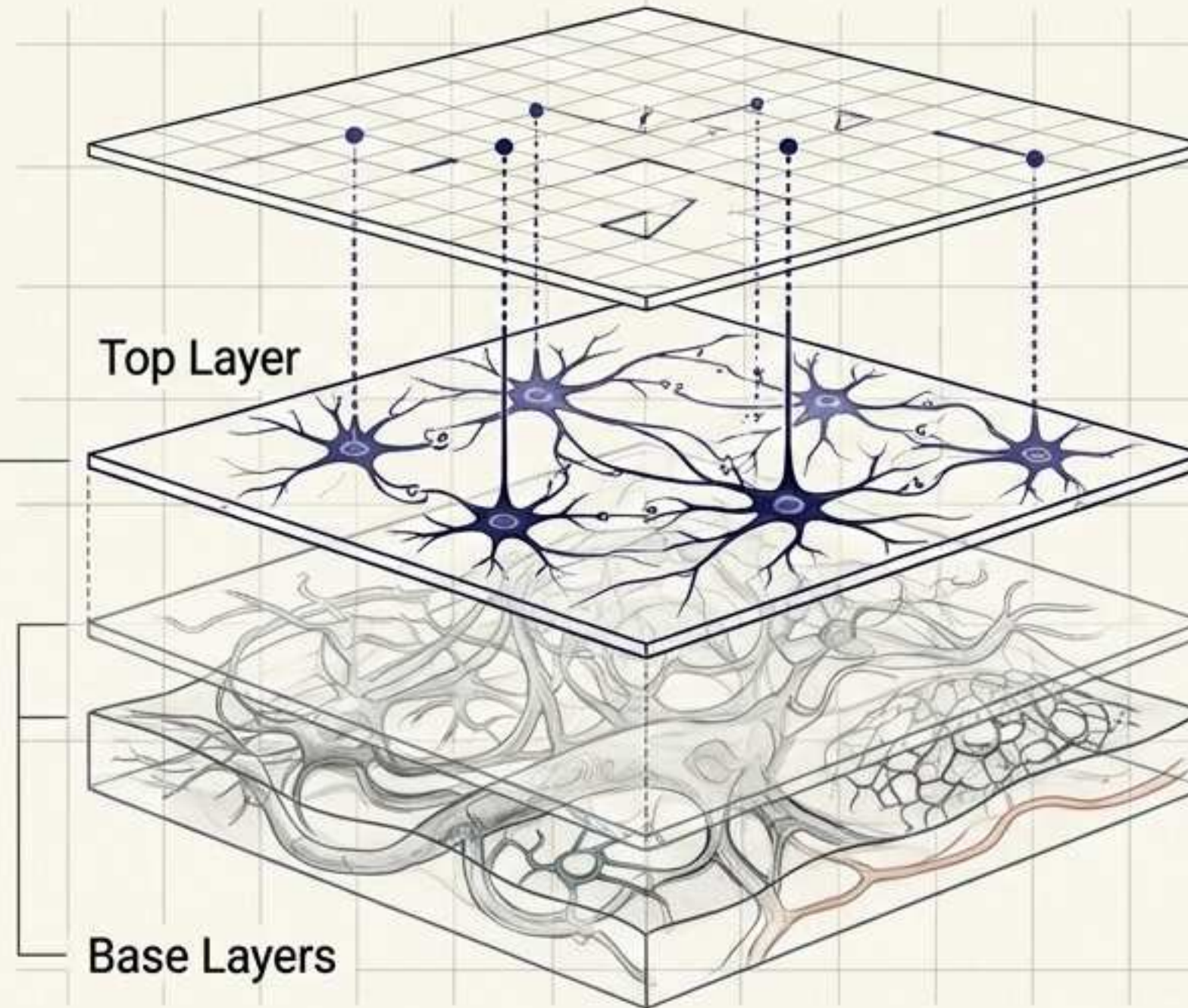
Standard projections (π_N) discard slow substrate variables that are causally relevant to future trajectories.

They artificially flatten reality.

Axiom:

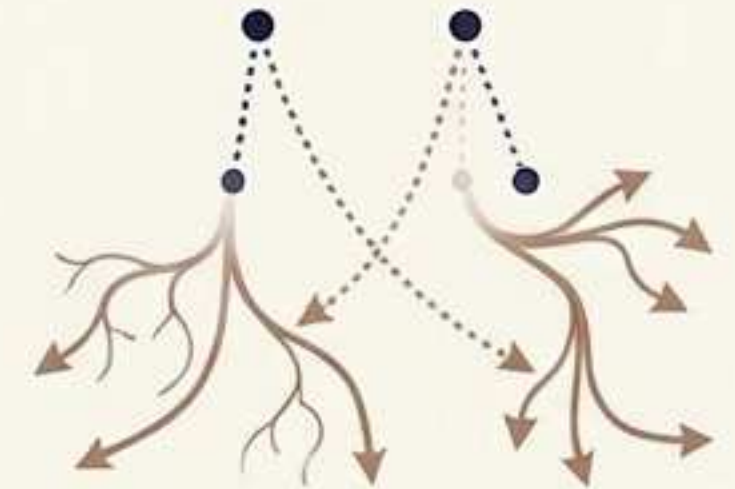
$$P(X) = \pi(X \mid S)$$

where S (substrate) is ignored.



Admissibility Distortion

Two states can look identical in the projection ($\pi(X_1) = \pi(X_2)$) but have completely different reachable futures ($R(X_1) \neq R(X_2)$).



Takeaway: The phenomena weren't absent; they were rendered invisible by the projection.

Repairing the Distortion: Two Empirical Mechanisms

	Mechanism 1: Substrate Remodeling	Mechanism 2: Trajectory Preservation
Action Taken	Keep projection fixed; change the substrate until it fits.	Keep substrate fixed; redesign the projection to track history.
Empirical Proof	Cohen et al. (Deep Neural Networks)	Busch et al. (T-PHATE fMRI Data)
Mechanism Result	Training remodels the network's weight substrate until the representational manifold geometry untangles visual categories. Learning is substrate repair.	Using an autocorrelation kernel preserves temporal adjacency, revealing neural event boundaries that point-in-time voxel projections destroy.

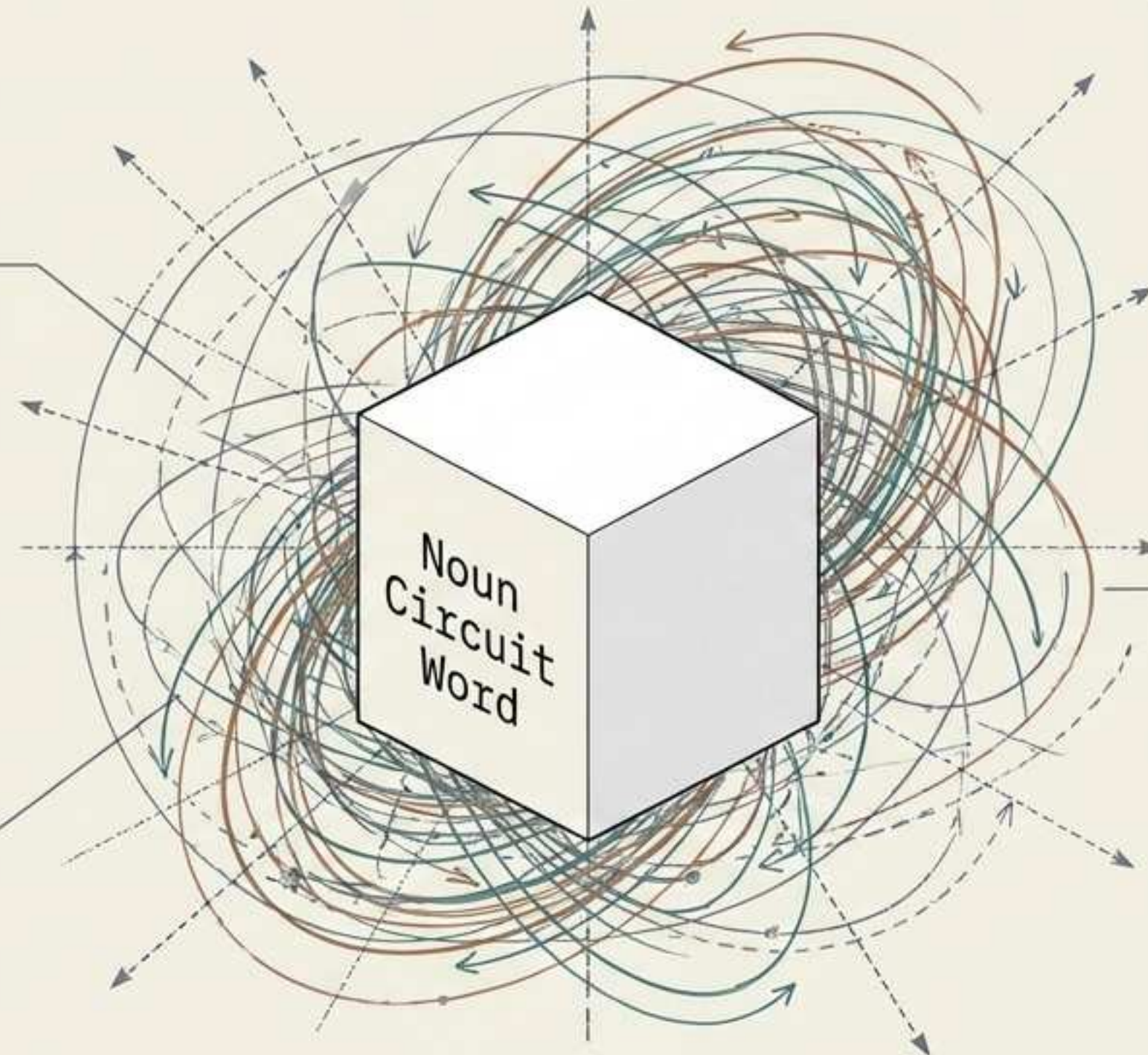
Pillar 3: Spherepop (The Noun Fallacy)

The Object-as-Collapse Theorem

Stable, discrete objects do not actually exist as primitive entities.

The Reality

Objects (O) are “collapsed residues” of recurring process histories ($\kappa(\gamma)$).



Exploded Noun

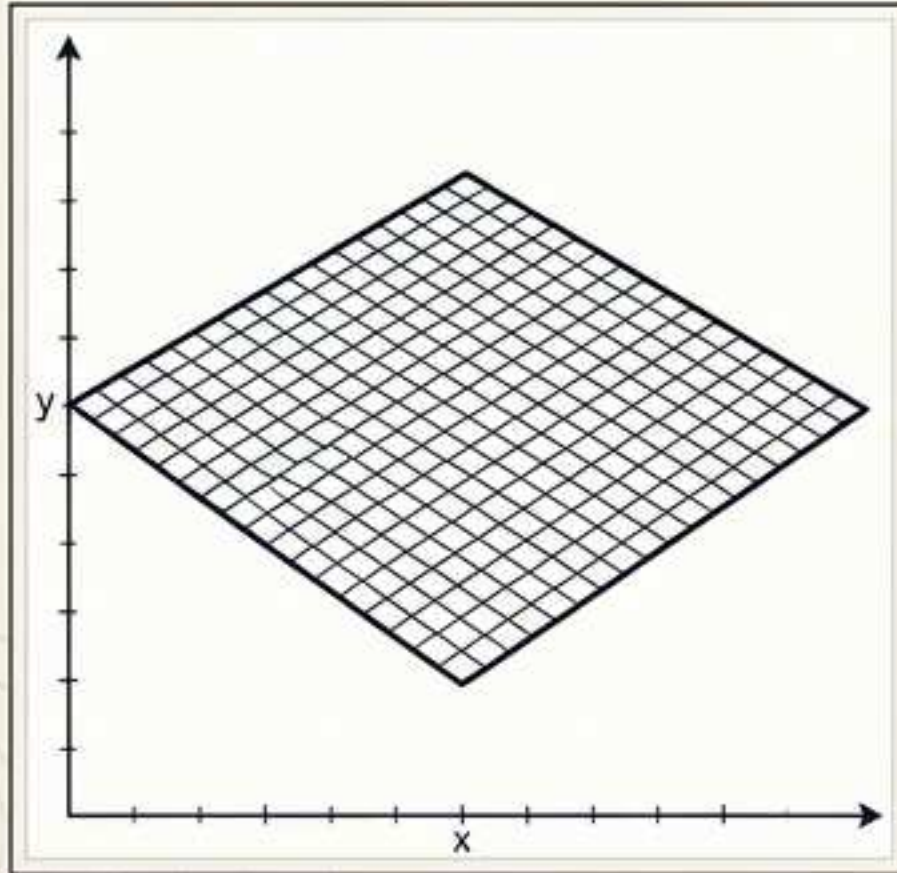
The Implication

If you zoom in on a noun—a brain circuit, a legal institution, a spoken word—it is just a frozen trajectory in the substrate.

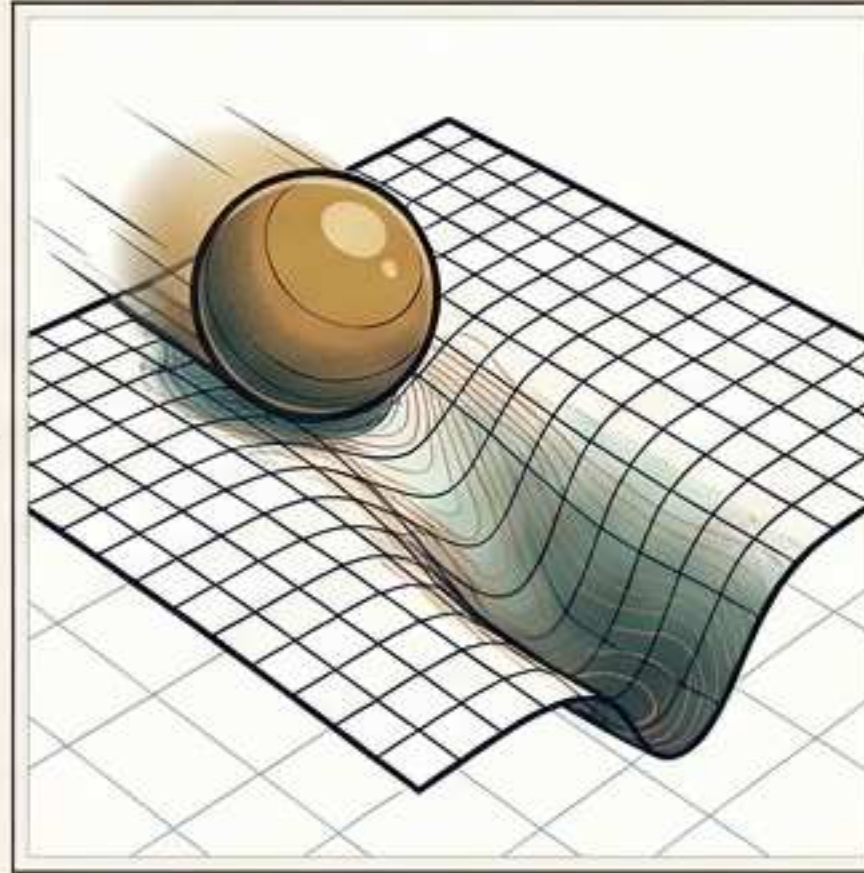
Memory as Substrate Deformation

Memory is not explicit state storage. It is the geometric deformation of the admissibility structure.

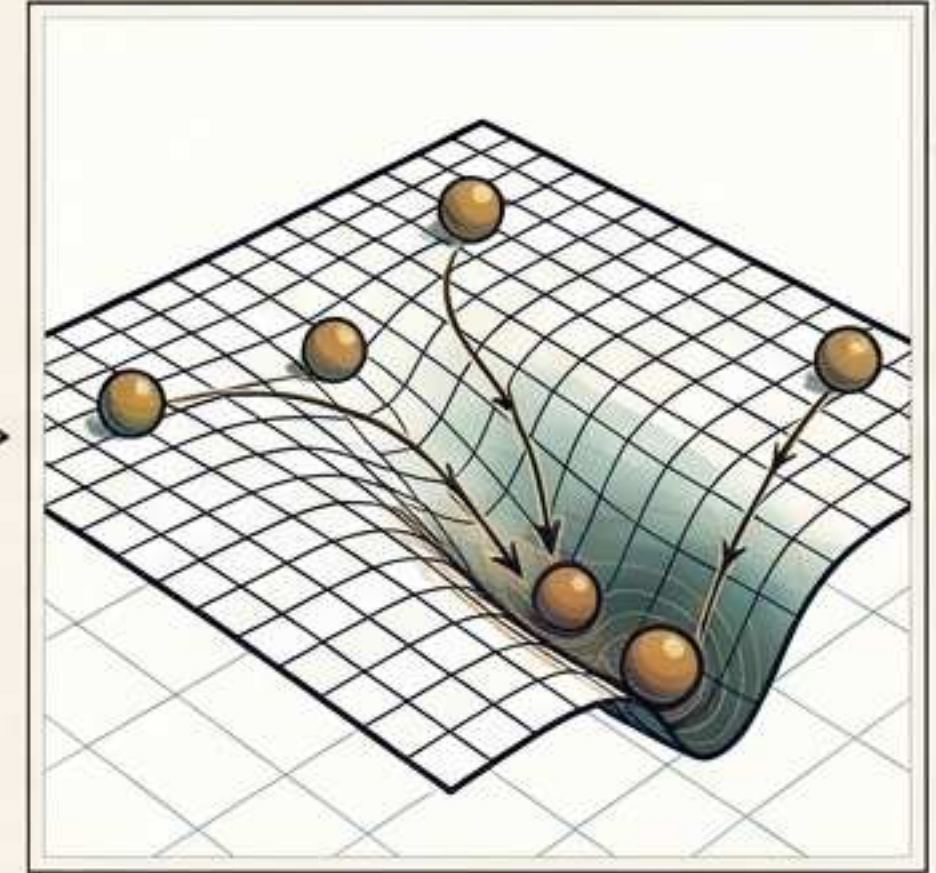
The Flat Substrate



Learning: Carving Attractors



Memory: Geometric Recall



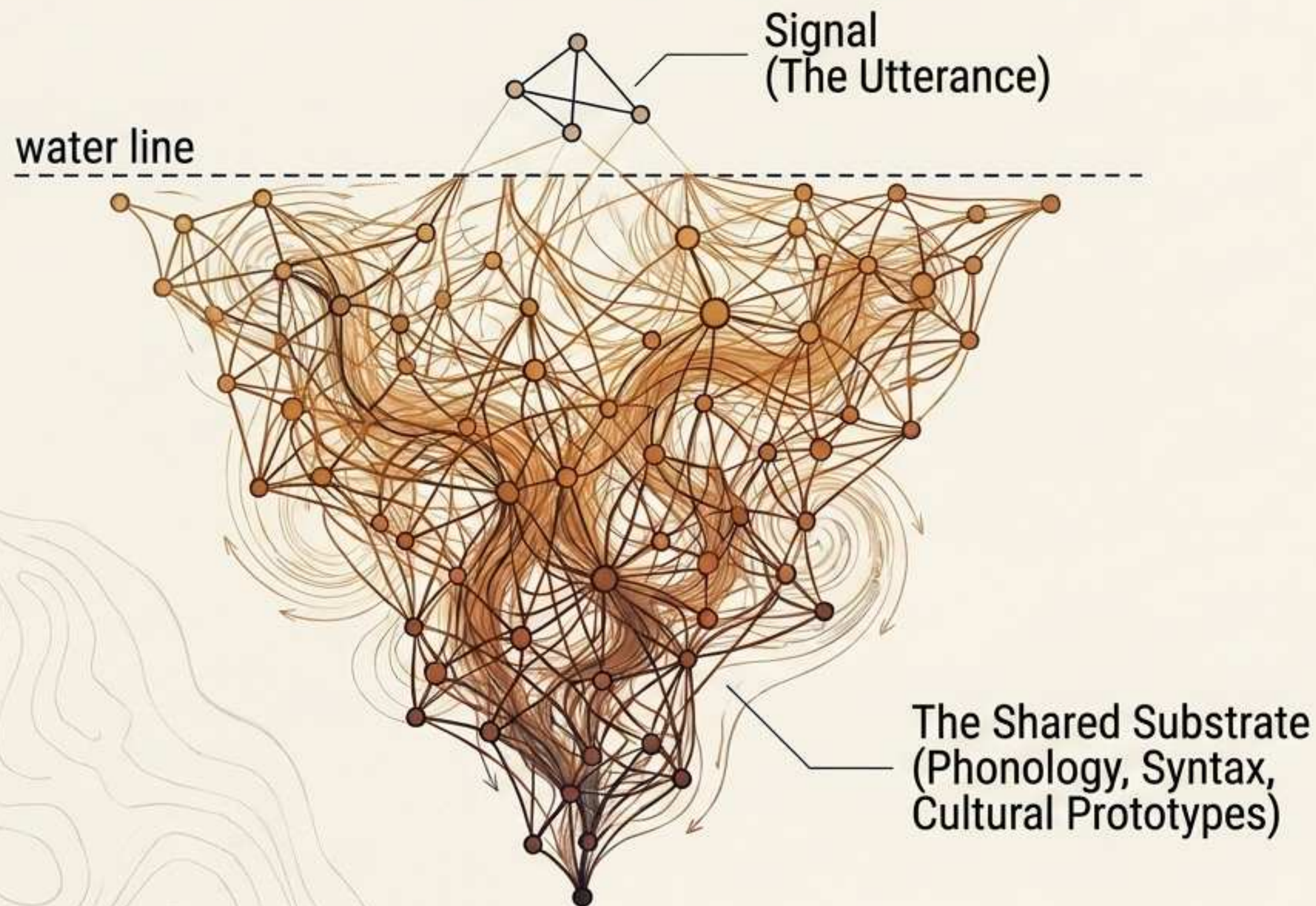
Learning & Forgetting

- **Learning:** Carving new attractor basins into the substrate.
- **Forgetting:** The collapse of those basins via substrate relaxation.

Catastrophic Forgetting (in AI)

Fast trajectories overwrite the weight substrate without the slow, regenerative maintenance ($R\Phi$) needed to preserve prior deformations.

Application 1: Language as Constraint Geometry



The Inversion

Language is not a signal encoding content. It is a sequence of distinction operations applied to a collectively maintained manifold.

Semantic Compression

We communicate efficiently because the shared substrate holds the context.

$$H_{\text{substrate}} \gg H_{\text{utterance}}$$

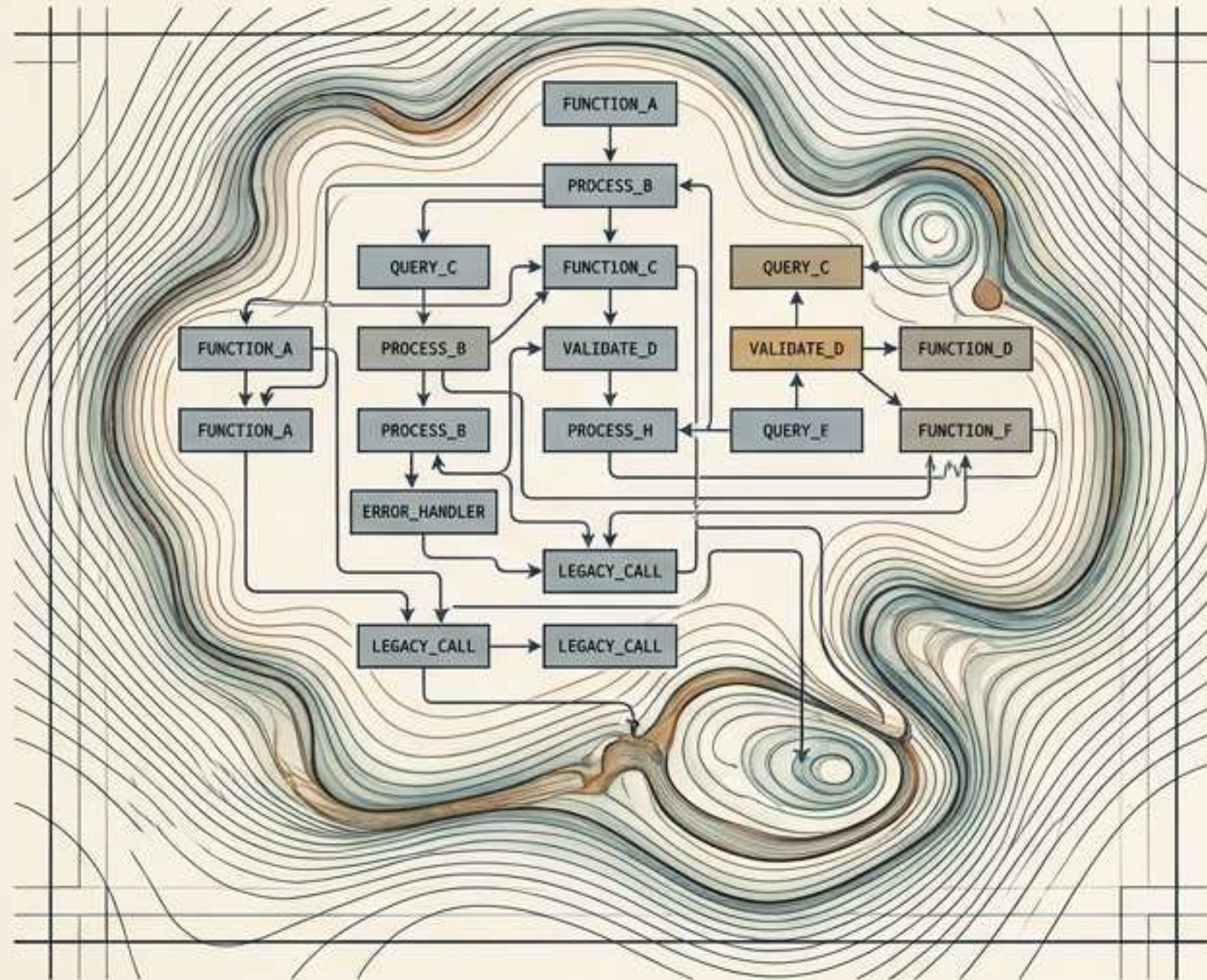
Words as Residue

The sentence doesn't carry the meaning; it activates a region of the shared substrate where the meaning is already latent.

Application 2: Technical Debt & Institutional Rigidity

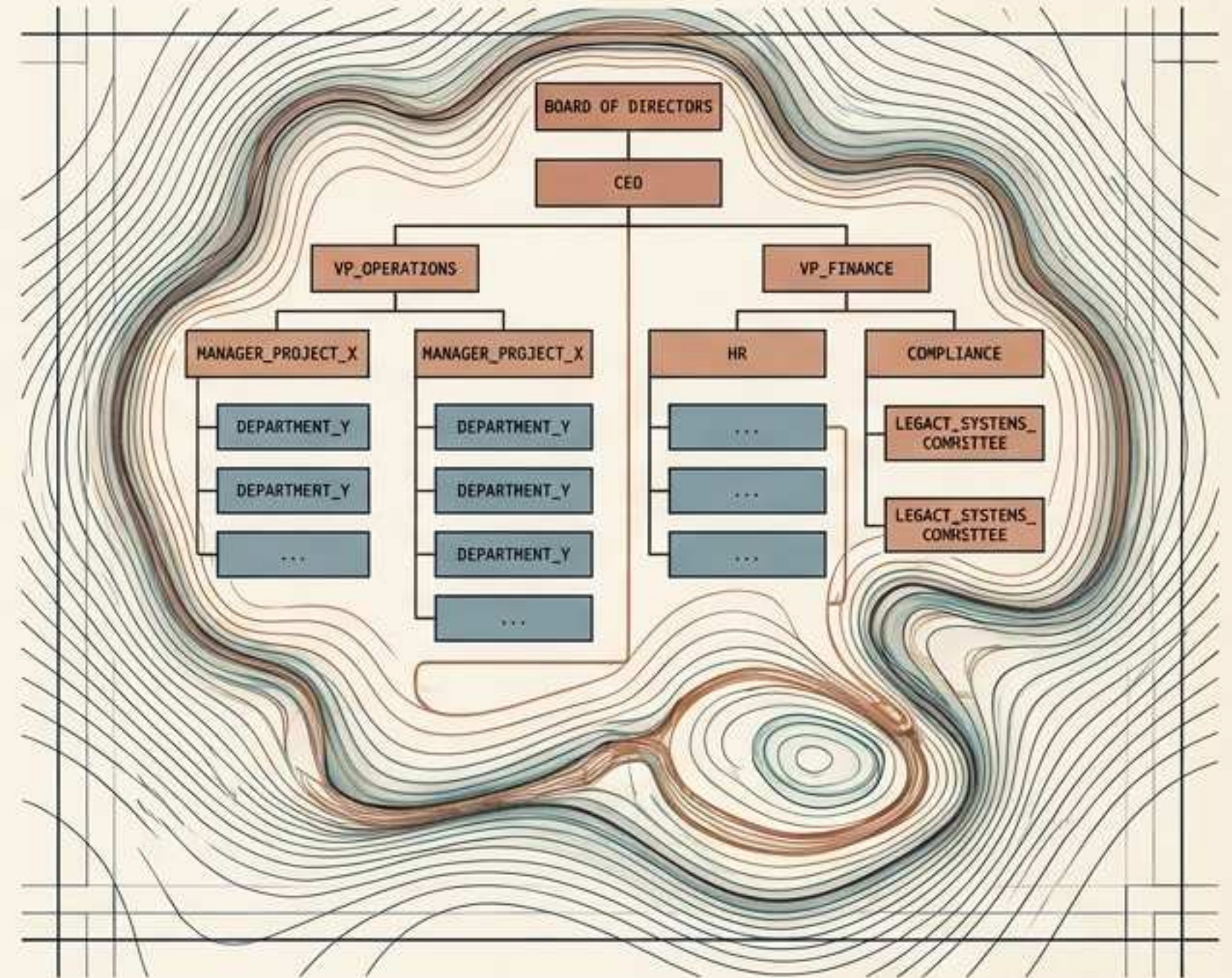
Institutional Path Dependence is not historical accident; it is geometric necessity.

Software



Technical debt is literal substrate deformation. Fast function calls warp the codebase, restricting future changes.

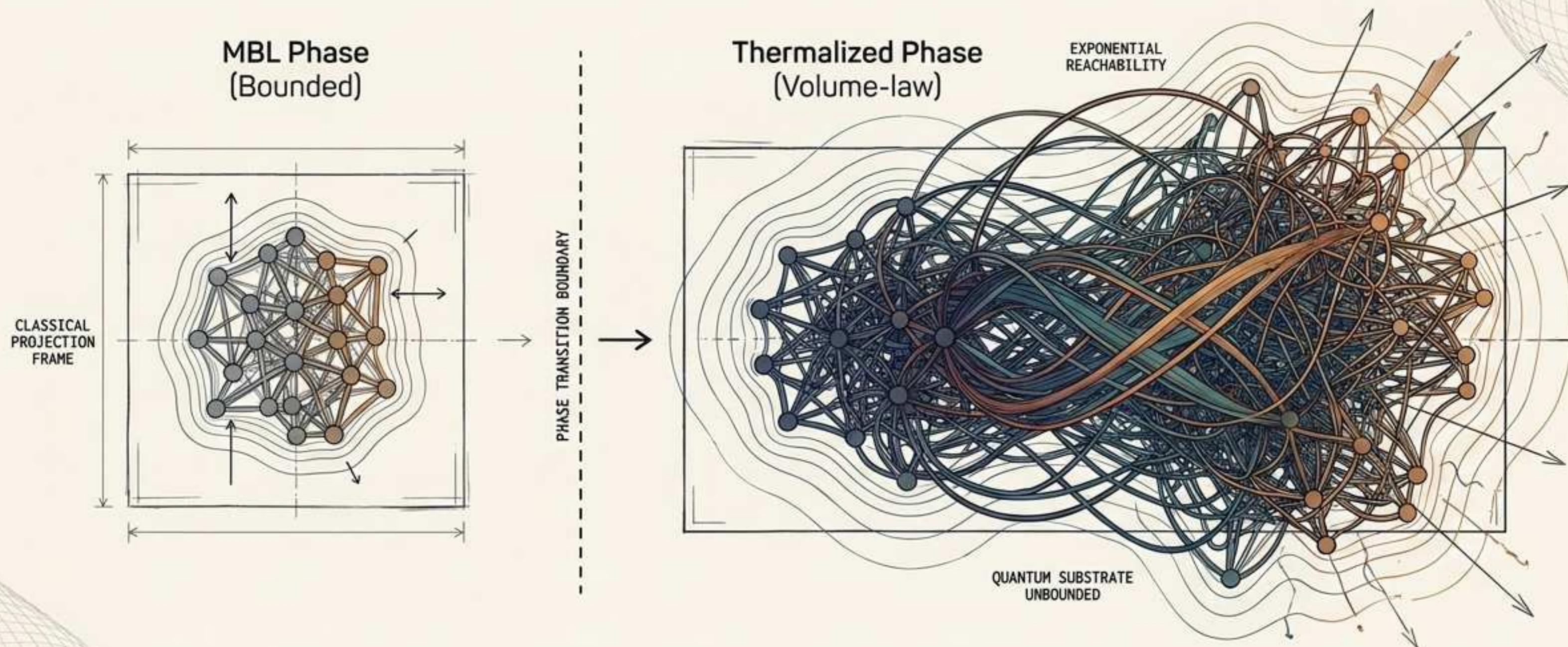
Institutions



Past decisions accumulate as structural constraints. Overcoming them requires explicit substrate repair, not just new choices.

Application 3: Quantum Computational Advantage

The Climax of CLIO.



The Climax of CLIO

Quantum advantage isn't just about "speed."
It is about reachability.

The Breakdown of Projection

In the thermalized phase, the quantum substrate's
reachable volume exceeds the expressive capacity
of any classically compressed description.

Result

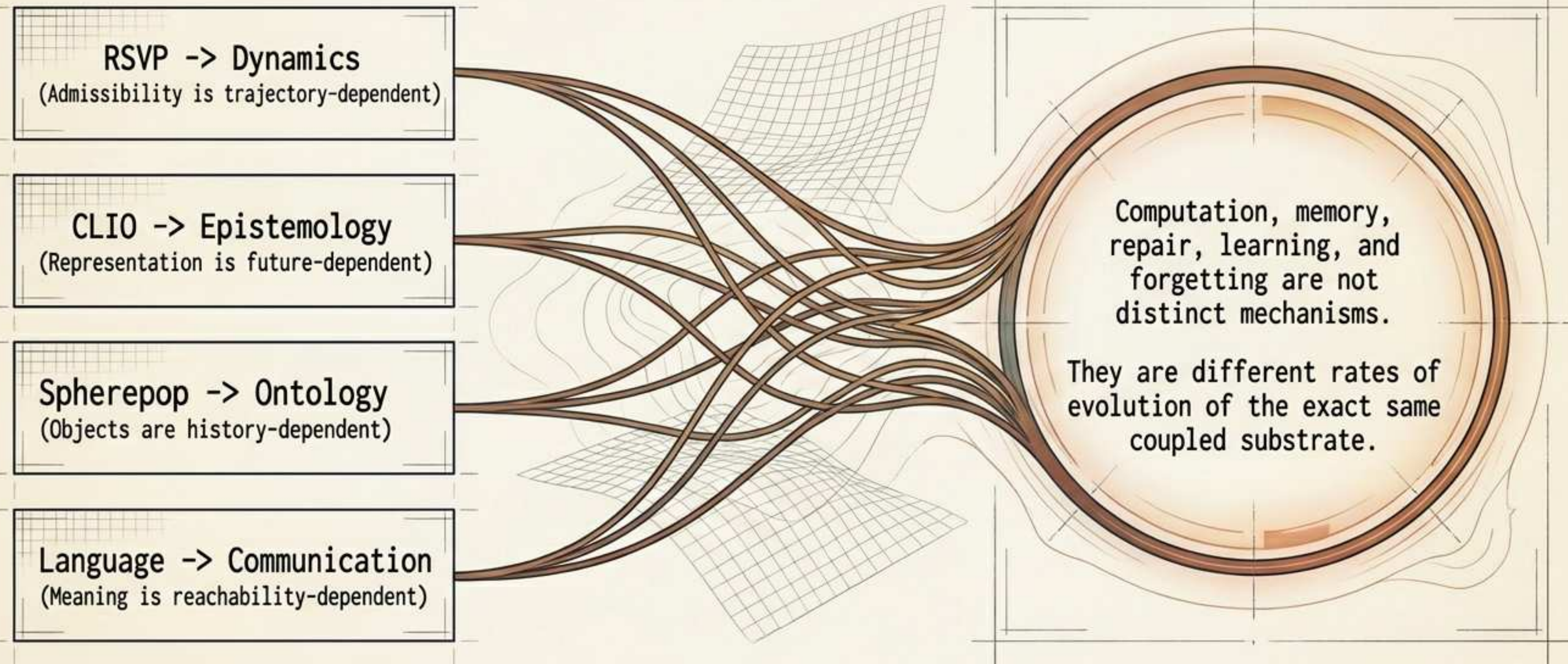
The quantum device succeeds because it doesn't
compress the dynamics—it IS the dynamics.

The Universal Substrate Matrix

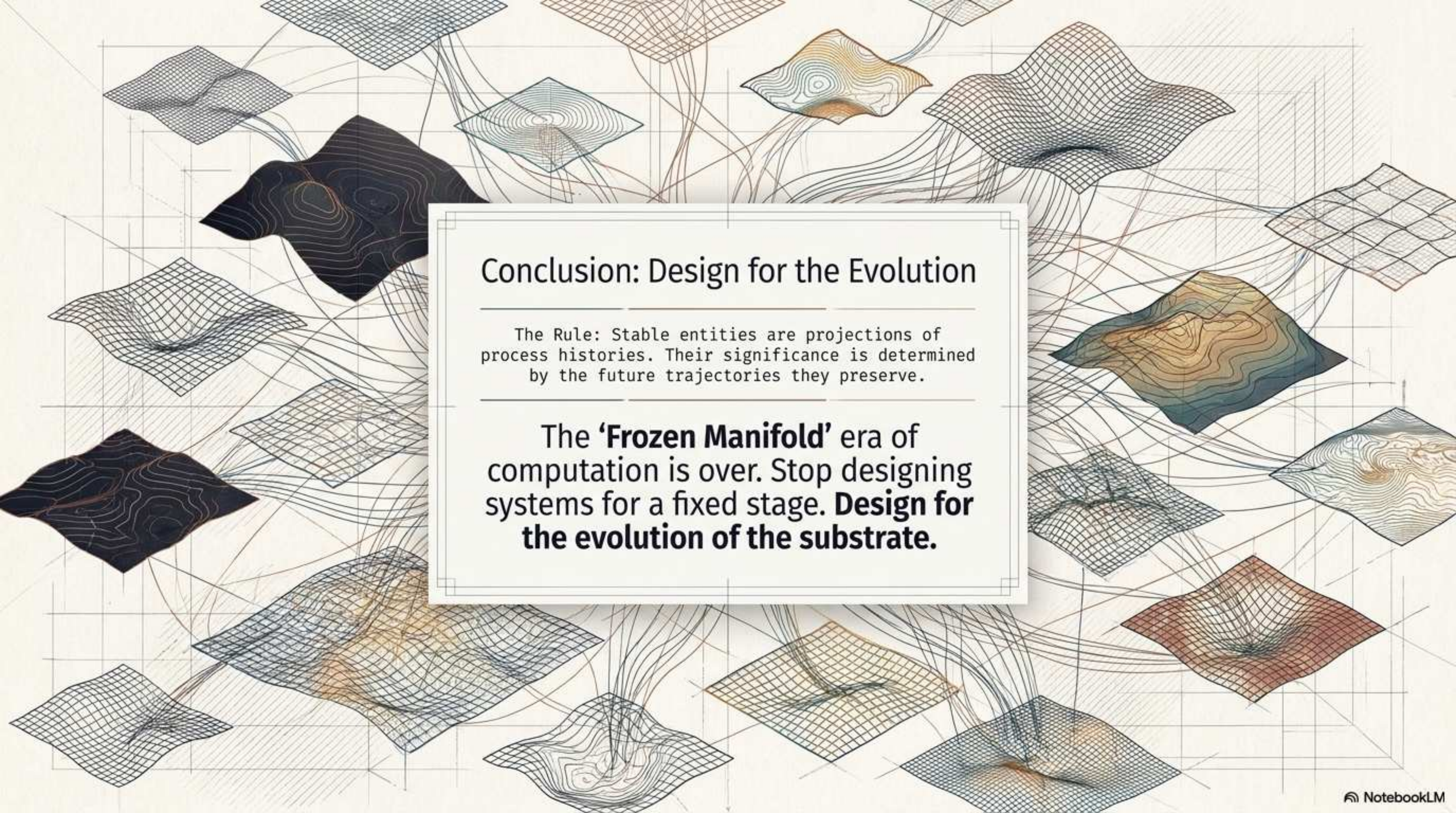
Domain	Fast Trajectory (x_t)	Slow Substrate (A_t)	Substrate Remodeling Force
Brain	Neural Spikes	Electrofluidic Field	Metabolic/Glymphatic Flow
Language	Utterances	Linguistic Community	Distinction Operations
Science	Experiments	Paradigm/Theory	Accumulated Anomalies
Institutions	Decisions	Rules/Norms	Precedent/Legislation
Economics	Transactions	Infrastructure	Investment/Wear
Software	Function Calls	Codebase Architecture	Technical Debt/Refactoring
Deep Networks	Classification Attempts	Weight Matrix	Gradient Descent

Across biological, social, and computational systems, the mathematical architecture is identical. Only the physical interpretation changes.

The Final Unification



The Fluid Blueprint: Across all systems, the mathematical architecture is invariant, only the physical interpretation and time scales change.

The background of the slide is a complex, abstract visualization. It features numerous wireframe surfaces, which are 3D representations of mathematical functions or data landscapes. These surfaces are rendered in various colors, including dark blue, light blue, orange, and brown. They are interconnected by a dense network of thin, curved lines that represent trajectories or paths. The overall effect is one of dynamic movement and interconnectedness, suggesting a complex system or process.

Conclusion: Design for the Evolution

The Rule: Stable entities are projections of process histories. Their significance is determined by the future trajectories they preserve.

The **‘Frozen Manifold’** era of computation is over. Stop designing systems for a fixed stage. **Design for the evolution of the substrate.**